Probability: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Basic terms

KEY POINTS

- Random experiment: outcome is one of known possible outcomes; sample space S = set of all outcomes; event = subset of S
- Impossible event ϕ: $P = 0$; sure event S: $P = 1$
- Mutually exclusive: $A_i \cap A_j = \phi$; exhaustive: $\cup A_i = S$; equally likely: same chance
- Mutually exclusive and exhaustive $\Rightarrow \sum P(A_i) = 1$; pairwise exclusive $\Rightarrow$ all exclusive

2. Classical definition and complement

Formulas

$$P(A) = \frac{n(A)}{n(S)}, \qquad P(A') = 1 - P(A)$$

KEY POINTS

- Outcomes must be equally likely: treat alike objects as distinct
- Count $n(A)$, $n(S)$ with P and C in the same way (both ordered or both unordered)
- "At least one", "not all", "solved", "hit": use $1 - P$(opposite)
- At least r: $1 - P(X \leq r - 1)$; at most r: $P(0) + \cdots + P(r)$

Watch out

Numerator and denominator counted differently (one ordered, one not)

3. Bounds on probabilities

Formulas

$$\max\{0, P(A) + P(B) - 1\} \leq P(A \cap B) \leq \min\{P(A), P(B)\}$$
$$\max\{P(A), P(B)\} \leq P(A \cup B) \leq \min\{1, P(A) + P(B)\}$$

KEY POINTS

- Every probability in $[0, 1]$; mutually exclusive events: sum ≤ 1; solve all inequalities together
- $P(A) = 0$ does not force $A = \phi$ in a general sample space

4. Odds

Formulas

$$\text{in favour} = \frac{P}{1 - P}, \qquad \text{against} = \frac{1 - P}{P}$$

KEY POINTS

- Odds in favour $p : q \Rightarrow P = \frac{p}{p + q}$; odds against $p : q \Rightarrow P = \frac{q}{p + q}$

5. Sample spaces of dice and coins

KEY POINTS

- n dice: 6^n ordered outcomes; n coins: 2^n; one die twice = two dice
- Two dice, ways of sum $2, \ldots, 12$: $1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$; doublet $\frac{1}{6}$
- Exactly r heads in n tosses: $\frac{{}^nC_r}{2^n}$

6. Venn diagram and union

Formulas

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = \sum P(A) - \sum P(A \cap B) + P(A \cap B \cap C)$$

KEY POINTS

- Only A: $P(A) - P(A \cap B)$; exactly one of A, B: $P(A) + P(B) - 2P(A \cap B)$; neither: $1 - P(A \cup B)$
- Three events: exactly two $= \sum P(A \cap B) - 3P(A \cap B \cap C)$; at least two $= \sum P(A \cap B) - 2P(A \cap B \cap C)$
- Exactly one of three $= \sum P(A) - 2\sum P(A \cap B) + 3P(A \cap B \cap C)$
- n events: $S_1 - S_2 + S_3 - \ldots$ (alternate signs); De Morgan: $(A \cup B)' = A' \cap B'$

STEPS

1. Write unknown region probabilities $a, b, c, \ldots$
2. Turn every given value into an equation; total $= 1$
3. Solve and add the regions asked

7. Conditional probability

Formulas

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{n(A \cap B)}{n(B)}$$

KEY POINTS

- The given event (after the bar) is the new sample space
- $P(A'|B) = 1 - P(A|B)$; $A \subseteq B$: $P(B|A) = 1$, $P(A|B) = \dfrac{P(A)}{P(B)}$; disjoint: $P(A|B) = 0$
- $\dfrac{P(A|B)}{P(B|A)} = \dfrac{P(A)}{P(B)}$

Watch out

$P(A|B') \neq 1 - P(A|B)$ and $P(A|B) \neq P(B|A)$

8. Multiplication theorem

Formulas

$$P(A \cap B) = P(A)P(B|A), \quad P(A_1A_2A_3) = P(A_1)P(A_2|A_1)P(A_3|A_1A_2)$$

KEY POINTS

- One by one without replacement: bag shrinks after each draw; with replacement: same bag
- Order not asked: ${}^nC_r$ method $= \dfrac{\text{favourable selections}}{{}^nC_r}$
- One-by-one for "one of each": multiply by the number of orders

9. Independent events

Formulas

$$P(A \cap B) = P(A)P(B); \qquad P(A \cup B) = 1 - P(A')P(B')$$

KEY POINTS

- Then A, B'; A', B; A', B' are independent
- Three events: 3 pairwise conditions and $P(ABC) = P(A)P(B)P(C)$
- Venn regions are products: only $A = p_1q_2$, exactly one $= p_1 + p_2 - 2p_1p_2$
- At least one of n: $1 - q_1q_2 \ldots q_n$

Watch out

Mutually exclusive with non-zero probabilities $\Rightarrow$ NOT independent

10. Repeated trials (infinite GP)

Formulas

$$P(X=k)=q^{k-1}p, \quad P(X>k)=q^k, \quad P(\text{1st of 2 wins})=\frac{p}{1-q^2}$$

KEY POINTS

- Three players in turn: first wins $\dfrac{p}{1-q^3}$
- Short method: $x+qx+q^2x=1$ type equations for the players' shares
- Number of tosses until first success: mean $\dfrac{1}{p}$

11. Total probability and Bayes

Formulas

$$P(A)=\sum P(B_i)P(A|B_i), \qquad P(B_i|A)=\frac{P(B_i)P(A|B_i)}{\sum P(B_j)P(A|B_j)}$$

STEPS

1. List the cases B_i (bags, coins, machines), mutually exclusive and exhaustive
2. Write $P(B_i)$ and $P(A|B_i)$ (tree diagram)
3. Effect asked: add the paths (total probability)
4. Cause asked after effect given: one path ÷ all paths (Bayes)
5. Next event after an observation: extended Bayes $\dfrac{\sum P(B_i)P(A\cap C|B_i)}{\sum P(B_i)P(A|B_i)}$

KEY POINTS

- Unknown composition ("all numbers equally likely"): each possible content is a case B_i

12. Binomial distribution

Formulas

$$P(X=r)={}^nC_r p^r q^{n-r}, \qquad \text{mean}=np, \ \text{variance}=npq$$

KEY POINTS

- Bernoulli trials: finite, independent, two outcomes, same p
- $\dfrac{P(X=r)}{P(X=r-1)}=\dfrac{n-r+1}{r}\cdot\dfrac{p}{q}$ (use for equal or AP conditions)
- Most probable r: $(n+1)p-1\le r\le(n+1)p$
- Least n for $P(\text{at least one})\ge k$: $q^n\le 1-k$
- $\dfrac{\text{variance}}{\text{mean}}=q<1$

Watch out

Drawing without replacement is not binomial

13. Tree diagram

KEY POINTS

- One level per stage; branch = conditional probability given the path so far
- Path probability = product; required probability = sum of favourable paths
- Branches from one node add to 1 (check)

14. Geometrical probability

Formulas

$$P = \frac{\text{favourable length/area/volume}}{\text{total length/area/volume}}$$

KEY POINTS

- Two real numbers in $[0, a]$: square of side a; $|x - y| \le d$ is the band between two lines

15. Random variable: mean and variance

Formulas

$$\sum p_i = 1, \qquad \mu = \sum p_i x_i, \qquad \sigma^2 = \sum p_i x_i^2 - \mu^2$$

KEY POINTS

- Unknown k in a table: $\sum p_i = 1$ and every $p_i \ge 0$ (reject the other root)
- Number of defectives in a sample without replacement: $P(X = r) = \dfrac{{}^{a}C_r \, {}^{b}C_{m-r}}{{}^{a+b}C_m}$
- Expectation of a game $= \sum p_i M_i$; $E(aX + b) = a\mu + b$, $\text{Var}(aX + b) = a^2\sigma^2$

16. Units digit of a product

Formulas

$$1, 3, 7, 9: \ \frac{4^n}{10^n}; \quad 2, 4, 6, 8: \ \frac{8^n - 4^n}{10^n}; \quad 5: \ \frac{5^n - 4^n}{10^n}; \quad 0: \ \frac{10^n - 8^n - 5^n + 4^n}{10^n}$$

KEY POINTS

- Only last digits matter; each digit 0 to 9 has probability $\dfrac{1}{10}$

17. Common mistakes checklist

STEPS

1. Are all outcomes equally likely? (distinct objects, ordered pairs for dice)
2. With or without replacement?
3. Which event is given? (conditional, Bayes)
4. Independent or mutually exclusive? Never mix them
5. Three events independent: check all four conditions
6. Probabilities found must lie in $[0, 1]$; reject other roots
7. Binomial needs independent trials with the same p